



Guidelines

Insulin: Adult Variable Rate Intravenous

Reference #: HRM003

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy
Out of Scope	This guideline is NOT suitable for the initial management of Diabetic Ketoacidosis (DKA) and Hyperosmolar Hyperglycaemic State (HHS). Separate guidelines are available. The IV Insulin management chart is NOT to be used for insulin in the treatment of hyperkalaemia. Insulin is charted on the standard Fluid Therapy & Intravenous Additive Orders chart. For more information refer to: SCGH Hyperkalaemia Guideline SCGOPHCG Nursing Practice Guideline No.4 Intravenous Therapy	

Introduction

This guideline provides hospital specific guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the prescribing, administration, storage or handling of adult variable rate intravenous insulin within SCGOPHCG. This document supports and complies with related legislation, policy and other related documents [here](#).

HIGH RISK MEDICINE

All management must be recorded on the Adult Variable Rate Intravenous Insulin Management Record (chart), see [Appendix 4](#). This document is designed to allow both prescriptions and relevant observations to be recorded together.

1.0 General Information^{1,2,3,4}

Insulin is a naturally produced hormone secreted by the pancreas. Insulin is required by the cells of the body in order for them to remove and utilise glucose from the blood. Insulin causes cells in the liver, muscle and fat tissues to absorb glucose and convert it to glycogen that can be stored. Additionally, insulin prevents the use of fat as an energy source. Administration of insulin results in lower blood glucose levels (BGLs).

Insulin is required for mandatory background (basal) and post prandial (bolus) blood glucose level (BGL) control. Insulin is considered a high-risk medication as errors in therapy can result in serious harm and can be fatal. Actrapid® (insulin human neutral) is a short acting insulin preparation; typical onset of action is within 30 minutes, peak effect after 2 to 3 hours and total duration of action of 6 to 8 hours.

Fasting before surgery or diagnostic procedures can present a clinical risk in people on insulin or oral hypoglycaemic agents. Most patients with Type 1 or Type 2 diabetes admitted to hospital can be treated with subcutaneous insulin. In some circumstances, medical practitioners may decide to use intravenous (IV) insulin therapy as an alternative.

2.0 Availability^{3,5}

- Actrapid ® (Insulin Human Neutral) 100 unit/mL, 10 mL vial.

3.0 Indications^{1,2}

1. Type 1 and unstable Type 2 insulin deficient patients with diabetes mellitus while fasting or unable to tolerate oral diet and fluids i.e. nil by mouth (NBM).
2. Unstable Type 2 patients with diabetes who are fasting or unable to tolerate oral medication and diet **without** evidence of [Diabetic Ketoacidosis \(DKA\)](#) and [Hyperosmolar Hyperglycaemic State \(HHS\)](#).
3. **Follow-on (late)** management of patients with DKA or HHS who are not yet eating and drinking normally.

4.0 Contraindications^{3,4}

- Hypoglycaemia – BGL (mmol/L) ≤ 3.9

5.0 Precautions^{3,4}

- Hypokalaemia – administration of insulin reduces extracellular potassium by facilitating a shift of potassium into cells.
- Hypersensitivity – if a patient has a documented allergy to insulin discuss with endocrinology team for further advice.
- Acute trauma or illness– insulin requirements may increase. Rate of infusion will need to be adjusted accordingly.
- Concurrent Corticosteroid therapy – Insulin requirements may increase whilst on corticosteroids and decrease upon their cessation. Rate of infusion will need to be adjusted accordingly.
- Total Parental Nutrition (TPN) - TPN may contain insulin and will therefore impact on the patient's BGL control. Although no rate adjustment is required for concurrent IV insulin and TPN, TPN with insulin may result in lower rate requirements. Patients receiving TPN therapy do not require an intravenous glucose infusion to be co-administered.

6.0 Dosage & Administration

- Insulin and glucose for the above indications must be prescribed on the Adult Variable Rate Intravenous Insulin Management Record.
- Standard dilution: 50 units of Actrapid ® (Insulin Human Neutral) in **50mL** sodium chloride 0.9%.
- Note: a 50mL sodium chloride 0.9% (Baxter) bag will contain a small amount of overage however should not be considered significant.
- There is no need to discard any of the insulin infusion before administration to the patient. Discarding is considered unnecessary as long as the individual is being monitored, as the amount of insulin binding to the priming tubing is negligible

Patients established on a subcutaneous **basal insulin** dose must continue basal insulin (long-acting insulin e.g. Optisulin®, Levemir®) dosing during intravenous insulin administration, even if fasting, to avoid hyperglycaemia and ketosis.

Patients established on subcutaneous ultra-long acting and ultra-short acting combination therapy with **Ryzodeg®** (insulin degludec/aspart) should have this withheld for the duration of their insulin/dextrose infusion. Seek Endocrinology advice before ceasing insulin infusion and restarting usual subcutaneous insulin dosing for these patients.⁷

- Prescribe the basal insulin on the Insulin Subcutaneous Order and Blood Glucose Record (See [Appendix 4](#)). See [Appendix 1](#) for a list of basal insulin available.

Glucose 10% must always run concurrently with IV insulin infusion.

Exception: Patients receiving TPN therapy do not require concurrent intravenous glucose infusion as TPN contains required glucose.

- Unopposed insulin administration carries a high risk of severe hypoglycaemia.

6.1 **Setting up the insulin and glucose 10% infusion**

- Administration must be via an infusion pump.
- Clearly label both the insulin infusion and the Glucose 10% infusion.
- Insulin and glucose 10% must be administered via two separate pumps. They can be administered through the same intravenous cannula via a two- or three-way cannula extension (Chooks foot).
- Refer to [Appendix 2](#) for further illustration/ information.

6.2 **Starting rate**

- A starting rate of infusion **must be indicated by the Medical Team** on the IV insulin chart.
- A suggested starting rate is 2 mL/hr (2 units Actrapid ® Insulin per hour) however consideration must be made to the patient's relative daily insulin requirements or insulin resistance and a higher rate may be indicated.
- For patients converting from DKA or HHS management: the infusion rate will need to be manually changed/indicated on the variable rate IV insulin management chart to the current equivalent dose administered as part of the DKA/HHS guideline.
- Please contact the Endocrinology Registrar via switch board if guidance is needed.

6.3 **Recording management on the “Adult Variable IV Insulin Management Record”**

- Nursing staff must record the start and stop time of the insulin and glucose infusion. Once the prescriber has completed the prescriptions appropriately, nursing staff may adjust insulin infusion rate after BGL's as per the Insulin Infusion Nomogram located on the front of the variable rate IV insulin management chart.
- When changing the rate, two nurses are required to check rate and sign insulin protocol chart.
- When halving rates, if needed, round down to one decimal place (e.g. 0.25mL/hour becomes 0.2mL/hour OR 4.25mL/hour becomes 4.2mL/hour). This is not necessary when using a syringe driver or the B-Braun Infusomat Space volumetric pumps.

6.4 **Monitoring**

- Blood glucose levels (BGL) are to be tested (ward blood glucose meter) hourly for the first 4 hours.
- If the insulin infusion rate remains unchanged after this period of time decrease BGL monitoring to 2-hourly for 12 hours, then consider 3-hourly monitoring if stable or as directed by the doctor.
- Revert to hourly monitoring if infusion rate changed.
- Monitor BGL hourly for 2 hours after stopping the insulin infusion, then as per routine monitoring outlined in [SCGOPHCG NPG No 2 Diabetes Patient Management](#).
- Ketones – monitor as per the [SCGOPHCG NPG No 2 Diabetes Patient Management](#) or as clinically indicated by the treating team.
- Check U&E's 6 hours after starting the infusion and then at least once daily thereafter.
- Potassium and additional fluid requirements to be decided by the prescribing doctor. These must be reviewed on a regular basis (at least daily). Fluid therapy additives e.g. potassium chloride in sodium chloride will be prescribed on a standard Fluid Therapy & Intravenous Additive Orders.
- If administering >10mmol per hour of potassium chloride **cardiac monitoring** must be in place; refer to the [SCGH Intravenous Potassium Supplement Guideline](#) and [SCGOPHCG High Risk Medications Policy](#).
- Seek Endocrinology team advice if regimen is failing to control patient's BGLs.

INSULIN INFUSION NOMOGRAM	
BGL (mmol/L)	Action
≥ 15 (15 or greater than)	Give 4mL (4 units) bolus of insulin infusion AND increase insulin infusion rate by 1mL/hr (= 1 unit/hr). Inform medical team and test ketones.
10 to 14.9	Increase insulin rate by 0.5mL/hr (= 0.5 unit/hr).
6 to 9.9	Maintain insulin infusion rate.
4 to 5.9	Halve the insulin infusion rate.
≤ 3.9 (3.9 or less)	Stop the insulin infusion (continue glucose infusion). Give 100mL of 10% glucose IV. Recheck BGLs in 15 minutes. Refer to the Subcutaneous Insulin Order and Blood Glucose Monitoring chart () for further advice on the management of hypoglycaemia. Inform medical team. Recommence insulin infusion at half the last tolerated rate when BGL ≥ 6.

6.5 Ceasing IV insulin therapy

- Stopping the infusion must only be determined and ordered by the treating medical team. As a guide the infusion should only stop once the patient is tolerating oral diet and fluids **normally**.
- For patients previously on **oral** hypoglycaemic (OHG) therapy:
 - Aim to stop IV insulin infusion early morning and give usual OHG's at the time of stopping.

6.5.1 For patients with known diabetes on multiple daily bolus or basal injections:

- Recommence usual bolus insulin with next meal.
- Cease IV insulin infusion 30 mins after bolus injection.
- *Note: if no basal insulin has been given for >24 hours aim to convert to subcutaneous insulin with evening meal and give usual basal dose **or***

convert at breakfast with half the usual basal dose and then give remainder of dose with evening meal.

6.5.2 For patients on CSI (insulin pump) therapy:

- seek advice from diabetes team / Endocrinology before recommencing pump.
- Cease IV insulin infusion 30 mins after recommencing pump.

6.5.3 For patients with newly diagnosed type 1 diabetes:

- Refer to [SCGH Adult Diabetic Ketoacidosis Guideline](#) or seek advice from diabetes team / Endocrinology before initiating subcutaneous therapy.

7.0 Practical Points

7.1 Drug interactions ^{3,4}

- Thiazolidinediones e.g. pioglitazone, rosiglitazone – increased risk of oedema and heart failure. Concurrent use of insulin and rosiglitazone is contraindicated; use combination with pioglitazone cautiously.

7.2 Adverse effects ^{3,4}

- Hypoglycaemia – most frequent and serious. Symptoms include sweating, hunger, faintness, palpitations, tremor, headache, visual disturbances and/ or altered mood.

7.3 Compatibility ⁶

- Compatible Fluids: Glucose 5%, glucose 10%, glucose 50%, Hartmann's via Y-site, Ringer's via Y-site, sodium chloride 0.9%.
- Compatible Drugs Y-Site: Amiodarone, caspofungin, ceftaroline, esmolol, ethanol, glyceryl trinitrate, milrinone, morphine sulfate, potassium chloride, sodium nitroprusside.
- Incompatibility with Drugs: aminophylline, cefoxitin, dopamine, glycopyrrolate, ketamine, labetalol, micafungin, noradrenaline, pentamidine, phentolamine, phenylephrine, piperacillin-tazobactam (EDTA Free), protamine, rocuronium.

7.4 Storage ⁶

- Each prepared solution can be stored or used for up to 24 hours at room temperature. Any remaining solution at this time should be discarded and a fresh solution prepared if the infusion is to continue.
- Actrapid® vials should be stored between 2 and 8° C.
- After first use, the vials may be kept at room temperature for a period of up to 4 weeks. The date of removal from refrigerated conditions must be indicated on the vial. Do not freeze.
- Protect from light; when not in use vials should be protected from excessive heat and light.
- The solution within the vial should appear clear and colourless; do not use if they appear uniformly white or cloudy.

7.5 Pregnancy and breastfeeding ⁶

- Pregnancy – Safe to use.
- Breastfeeding – Safe to use.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- NSQHS Standards: 1,4,6,8

Standards, procedures, guidelines:

- [SCGH Adult Diabetic Ketoacidosis Guideline](#)
- [SCGH Adult Hyperosmolar Hyperglycaemic State \(HHS\) Guideline](#)
- [SCGH Hyperkalaemia Guideline](#)
- [SCGOPHCG Nursing Practice Guideline No.2 Diabetes Patient Management](#)
- [SCGOPHCG Nursing Practice Guideline No.4 Intravenous Therapy](#)

Supporting documents

- SCGOPHCG Adult Variable Rate Intravenous Insulin Management Record
- SCGOPHCG Adult Diabetic Ketoacidosis Guidelines and Management Record
- SCGOPHCG Adult Hyperosmolar Hyperglycaemic State Guidelines and Management Record

References

1. eTG complete. Endocrinology 2016 Available from : <https://www-tg-org-au.qelibresources.health.wa.gov.au/>
2. Khan N, Ghali W, Cagiero E. Up to Date. Endocrinology and Diabetes. Perioperative management of blood glucose in adults with diabetes mellitus. February 2018. Available from <https://www-uptodate-com.qelibresources.health.wa.gov.au>
3. Mims Online. Full Product Information Actrapid ®. Available from: <https://www-mimsonline-com-au.qelibresources.health.wa.gov.au/>
4. Australian Medicine Handbook. AMH. Insulins Available from: <https://amhonline-amh-net-au.qelibresources.health.wa.gov.au/>
5. Internal correspondence – Department of Endocrinology & Diabetes – Dr Joey Kaye
6. Australian Injectable Drug Handbook 7th Edition. Insulin-Short Acting. Available from <https://aidh-hcn-com-au.qelibresources.health.wa.gov.au/>
7. Internal correspondence – Department of Endocrinology and Diabetes – Gloria Daniels CNC

Acknowledgments

Fiona Stanley Hospital Adult Intravenous Insulin Infusion Chart MR 835

Authorisation & Version Control

Policy Sponsor	Director of Clinical Services				
Policy Contact	Drugs and Therapeutics Committee Executive Officer (Senior Pharmacist)				
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Standards applicable:	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1. List of commonly available Insulins in Australia

Type	Brand Name	Nature
Ultra-short-acting (Peak at one hour, last 4 - 5 hours)		
Insulin lispro	Humalog ®	Analogue
Insulin aspart	NovoRapid ®	Analogue
Faster-acting insulin aspart	Fiasp®	Analogue
Insulin glulisine	Apidra ®	Analogue
Short-acting (Peak at 2 to 5 hours, last 6 – 8 hours)		
Neutral	Actrapid ®	Human
	Humulin R ®	Human
	Hypurin Neutral ®	Bovine
Long acting (16-24 hours)		
Isophane	Humulin NPH ®	Human
	Protaphane ®	Human
Long acting Mixed (16-24 hours)		
Long acting mixed with ultra-short acting	Novomix 30 ®	Analogue
	Humalog Mix 25 ®	Analogue
	Humalog Mix 50 ®	Analogue
Long acting (Basal)		
Insulin detemir	Levemir ®	Analogue
Insulin glargine	Optisulin ®*	Analogue
Insulin glargine	Toujeo ® *	Analogue
	* Note Optisulin ® and Toujeo® are not bioequivalent - seek specialist advice if converting patient.	

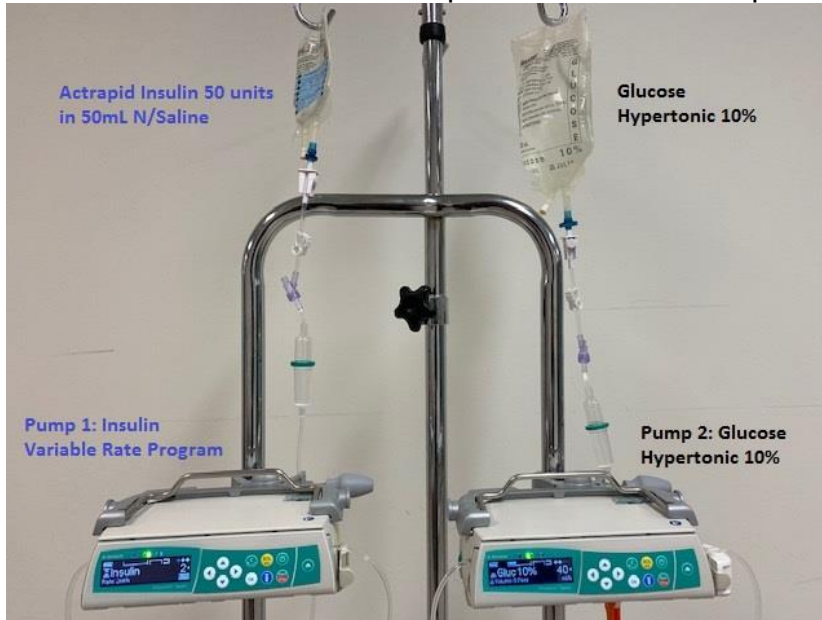
Not all insulin formulations available for initiation at SCGH/OPH – Please refer to [Formulary1](#) for prescribing restrictions.

Reference: Adapted from Australian Medicine Handbook. AMH. Insulins Available from: <https://amhonline-amh-net-au.qelibresources.health.wa.gov.au/>

Appendix 2. Illustration - Setting up the Insulin Glucose 10% Infusion

- Ensure that each pump is marked clearly.
- Pump 1 for Actrapid® in sodium chloride 0.9% infusion.
- Pump 2 for glucose infusion.
- Use the same cannula and connect to a chooks foot.
- See picture 1 for suggested B-Braun Infusomat Space Volumetric Pumps set up.

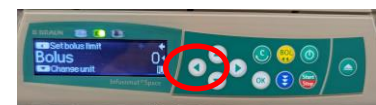
Picture 1: B-Braun Infusomat Space Volumetric Pumps



Created by G71 Nursing staff

Appendix 3. Administering an IV bolus from a B-Braun Infusomat Space Volumetric Pump

1. Ensure you are bolusing from the Insulin Infusion only.
2. Press the yellow bolus key.
3. Press the left arrow to select dose to bolus
4. The pump will choose by default a dose of 4 units
5. Press the yellow bolus key to initiate bolus
6. You will hear a tone, with a count indicating how much bolus has been administered, followed by a second tone to signal the completion of the bolus
7. The pump will resume at the previously selected infusion rate



BEEP



BEEP



Appendix 4. Picture of the Adult Variable Rate Intravenous Insulin Management Record

USE PATIENT LABEL WHEN AVAILABLE

SIR CHARLES GAIRDNER OSBORNE PARK HEALTH CARE GROUP ADULT VARIABLE RATE INTRAVENOUS INSULIN MANAGEMENT RECORD		URN: Family name: _____ Given name(s): _____ Gender: _____ DOB: 						
First prescriber to print patient name and check label correct:								
Ward: _____		ADVERSE DRUG REACTION LABEL						
Consultant: _____								
FOR FURTHER INFORMATION REFER TO: VARIABLE RATE INTRAVENOUS INSULIN GUIDELINE (ADULT)								
- This chart is not to be used for diabetic emergencies i.e. Diabetic Ketoacidosis (DKA) & Hyperosmolar Hyperglycaemic State (HHS), or for insulin in the treatment hyperkalaemia. - Unopposed insulin administration carries a high risk of severe hypoglycaemia and insulin infusions should ALWAYS be accompanied by co-administration of intravenous glucose (EXCEPTION: Patients receiving TPN therapy do not require an intravenous glucose infusion) - If specialist advice is required, contact the ENDOCRINOLOGY team via switch board - Continue to administer subcutaneous Basal Insulin (i.e. Optisulin / Levemir®) as prescribed on the Insulin Subcutaneous order and Blood Glucose record form.								
DATE	TIME	INSULIN INFUSION ORDER	STARTING INSULIN RATE	DOCTOR'S NAME (PRINT)	DOCTOR'S SIGNATURE	GIVEN BY	CHECKED BY	PHARM
		50 units Actrapid® in 50mL Sodium Chloride 0.9%	2 mL/hr (2 units/hr)					
		50 units Actrapid® in 50mL Sodium Chloride 0.9%	OR, Indicate alternative start rate if converting from DKA/HHS: _____ mL/hr					
DATE	TIME	GLUCOSE INFUSION ORDER	STARTING GLUCOSE RATE	DOCTOR'S NAME (PRINT)	DOCTOR'S SIGNATURE	GIVEN BY	CHECKED BY	PHARM
		10% glucose 500mL	40 mL/hr					
Note: Consider additional hydration as appropriate								
Prescriber to complete ☐ Prescriber to be contacted following each BGL test. Contact details: _____ ☐ Nursing staff to adjust dose based on nomogram below. ☐ When halving rates, round down to one decimal place (e.g. 4.25 ml/hour becomes 4.2 ml/hour). (Tick as appropriate) This is not necessary when using a syringe driver or the B-Braun Infusomat Space volumetric pumps.								
INSULIN INFUSION NOMOGRAM								
BGL (mmol/L)		Action						
≥15 (15 or greater than)		Give 4mL (4 unit) bolus of insulin infusion AND increase insulin infusion rate by 1 mL/hr (=1 unit/hr). Inform medical team and test ketones						
10 to 14.9		Increase insulin rate by 0.5mL/hr (=0.5 unit/hr)						
6 to 9.9		Maintain insulin infusion rate						
4 to 5.9		Halve the insulin infusion rate						
≤3.9 (3.9 or less)		Stop the insulin infusion (continue glucose infusion). Give 100mL of 10% glucose IV. Check BGL 15 minutely. Refer to Subcutaneous Insulin Order and Blood Glucose Monitoring Chart for further management of hypoglycaemia. Inform medical team. Recommence insulin infusion at half the last tolerated rate when BGL ≥8.						

ADULT VARIABLE RATE INTRAVENOUS
INSULIN MANAGEMENT RECORD

M100
03/21

MR826 /
MR(OPH)156A

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